

Assignment 1 – PS4 Defence Drone

Greedy Best-First Search (GBFS) and A* Search Analysis

1. Problem Statement

An autonomous UAV must navigate from Start node (0,0) to Goal node (6,7) while avoiding No-Fly Zones (N) and accounting for Weather Hazard regions (W). The objective is to evaluate Greedy Best-First Search (GBFS) using two heuristics and compare its performance with A* Search.

2. Environment Representation

Grid Size: 8×8. Symbols used: S=Start, E=Goal, .=Open Cell, W=Weather Hazard, N=No-Fly Zone. Movement is restricted to North, East, South and West.

3. PEAS Analysis

Performance: Reach the target with minimum risk and cost.

Environment: 8×8 grid world.

Actuators: Move North, East, South, West.

Sensors: Current position and neighboring cells.

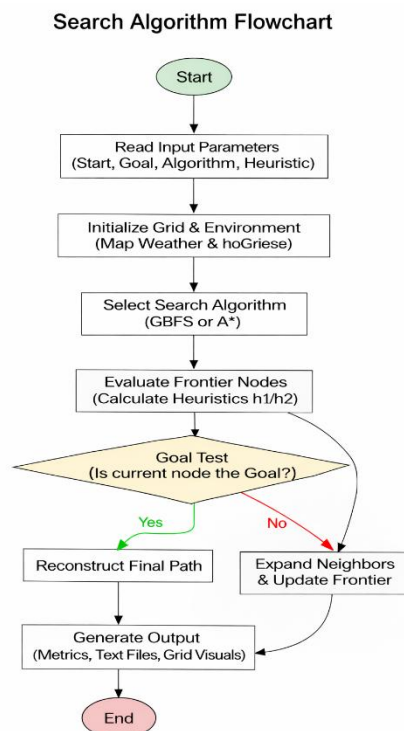
4. Heuristic Functions

h1: Euclidean Distance.

h2: Bounding-Box Risk Weighted Heuristic which combines Manhattan distance with average future path risk.

5. Execution Flow

Read Input → Validate Input → Build Grid → Select Search Algorithm → Expand Nodes → Compute Heuristic → Goal Test → Reconstruct Path → Generate Output.



6. Initial Environment and Planned Route

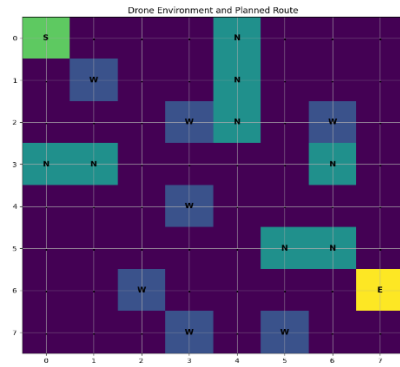
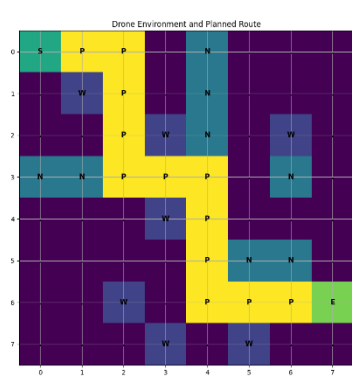
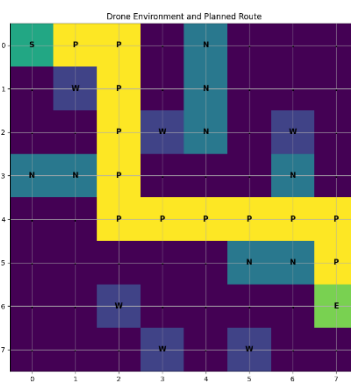
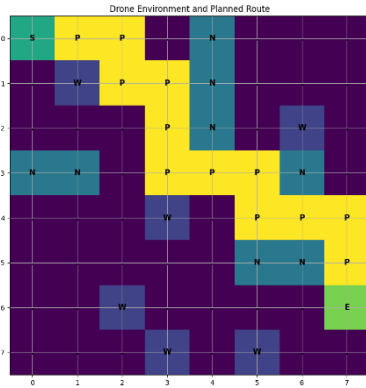


Figure 6 (a): Initial Environment and Planned Route shows the defense drone environment. The start node S is located at (0,0) and the goal node E at (6,7). No-Fly Zones (N) are treated as obstacles while Weather Hazard cells (W) have higher traversal cost.

7. Final Path Visualization



| GBFS-h1 Final Path (Cost = 33) Figure 7(a) | GBFS-h2 Final Path (Cost = 19) Figure 7(b) | A*-h1 Final Path (Cost = 16) Figure 7(c) |

8. Comparative Analysis

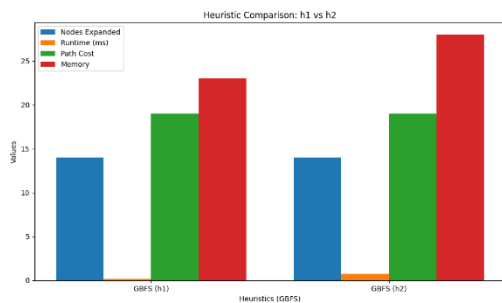
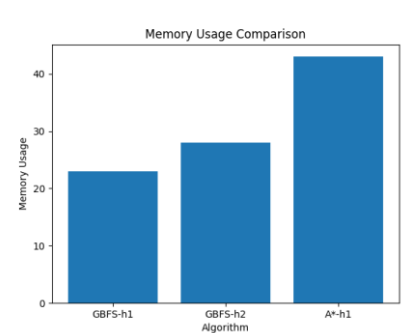
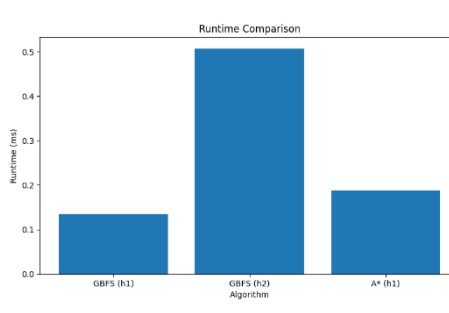
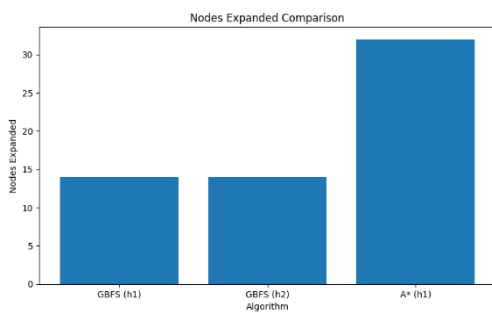


Fig 8 (a) - Nodes Expanded: GBFS expands fewer nodes for faster execution, while A* explores more nodes to ensure optimality. Fig 8 (b) – Runtime: GBFS-h1 has the lowest runtime due to greedy selection, whereas GBFS-h2 and A* require slightly more computation. Fig 8 (c) - Memory Usage: Memory usage increases from GBFS-h1 to A*, reflecting the additional data maintained for optimal path computation.

The comparative results demonstrate the impact of heuristic design and search strategy on path quality and search efficiency. GBFS-h1 expanded the fewest nodes (14) and achieved the fastest execution time by greedily selecting nodes with the smallest Euclidean distance to the goal. However, because Greedy Best-First Search does not consider the accumulated path cost, it was attracted toward the goal without accounting for hazardous terrain, resulting in traversal through a Weather Hazard cell and producing the highest path cost of 33. In contrast, GBFS-h2 employed the Bounding-Box Risk-Weighted Heuristic, which incorporates regional risk information into the search process. This enabled the algorithm to avoid high-risk areas more effectively, reducing the overall path cost to 19 while maintaining relatively efficient search behavior. A*-h1 produced the best overall solution with the minimum path cost of 16 by combining the actual cumulative path cost $g(n)$ with the Euclidean heuristic estimate $h(n)$. This balanced evaluation allowed the algorithm to avoid unnecessary hazard traversal and identify the lowest-cost feasible route. All three algorithms successfully avoided No-Fly Zones, demonstrating correct constraint handling. Overall, the results highlight the trade-off between search efficiency and solution optimality: GBFS variants achieve faster decisions through heuristic guidance alone, whereas A* incurs additional computation to guarantee a lower-cost and more optimal path.

9. Experimental Results

Metric	Description	GBFS-h1	GBFS-h2	A*-h1
Nodes Expanded	Total explored nodes	14	14	32
Runtime (ms)	Execution time	0.130388	0.636698	0.317236
Memory Usage	OPEN + CLOSED size	26	32	51
Total Path Cost	Sum of transition costs	33	19	16
Path Length	Total moves	13	13	13
Initial Heuristic Value	Start node heuristic value	9.22	29.95	9.22

The experimental results reveal that all three algorithms reached the goal using paths of identical length (13 moves), indicating that path length alone is not a sufficient indicator of solution quality in weighted environments. Despite producing routes with the same number of movements, the total path costs varied significantly due to differences in hazard avoidance and node-selection strategy. GBFS-h1 generated the highest path cost (33), showing that minimizing estimated distance to the goal does not necessarily minimize traversal cost. GBFS-h2 reduced the cost to 19 by incorporating regional risk information into its heuristic evaluation, demonstrating the benefit of risk-aware guidance. A*-h1 achieved the minimum path cost of 16 while maintaining the same path length, illustrating the effectiveness of combining accumulated cost and heuristic estimation. The memory usage trend (26 $\rightarrow$ 32 $\rightarrow$ 51) further highlights the trade-off between computational resources and solution quality, with A* requiring additional state information to guarantee an optimal result. The substantially larger initial heuristic value for h2 (29.95 versus 9.22) also indicates that the risk-weighted heuristic captures environmental difficulty more aggressively than pure Euclidean distance.

10. Complexity Analysis

Greedy Best-First Search (GBFS) uses a priority queue ordered by the heuristic value $h(n)$. Its time complexity is approximately $O((V+E)\log V)$ and space complexity is $O(V)$. In this experiment, GBFS expanded only 14 nodes, making it computationally efficient.

A* Search uses the evaluation function $f(n)=g(n)+h(n)$, combining path cost and heuristic information. Its time complexity is also $O((V+E)\log V)$ with $O(V)$ space complexity. Although A* expanded 32 nodes and used more memory, it generated the optimal path with the lowest cost.

Algorithm	Time Complexity	Space Complexity
GBFS-h1	$O((V+E) \log V)$	$O(V)$

GBFS-h2	$O((V+E) \log V)$	$O(V)$
A*-h1	$O((V+E) \log V)$	$O(V)$

11. Trap Analysis

No dead-end states, local minima, or infinite loops were encountered. All algorithms successfully reached the goal without requiring trap-recovery mechanisms.

12. Alternate Problem Modelling

The environment can alternatively be modelled as a graph, where each grid cell is represented as a node and adjacent traversable cells form edges. This representation improves scalability and supports advanced search algorithms such as Dijkstra's and A*, but requires additional memory to store graph structures. For the current 8×8 environment, the grid representation is simpler and more efficient.

13. Completeness and Optimality Comparison

Algorithm	Complete	Optimal
GBFS	Yes (for this finite grid)	No
A*	Yes	Yes

GBFS successfully reached the target in all experiments and is complete for this finite environment. However, it does not guarantee an optimal solution because the heuristic alone determines node selection. This is reflected in the higher path costs produced by GBFS-h1 (33) and GBFS-h2 (19). A* is both complete and optimal because it considers the accumulated path cost and heuristic estimate simultaneously, enabling it to find the minimum-cost path of 16.

14. Conclusion

All three search configurations successfully reached the target node. GBFS-h1 proved to be highly computationally efficient, achieving the fewest node expansions (14) and the lowest memory footprint (26). However, because the Euclidean distance ignores environmental obstacles, it guided the drone into a high-cost trajectory close to obstacle boundaries and through a Weather Hazard zone, accumulating a path cost of 33. GBFS-h2 expanded the same number of nodes (14) but significantly improved route quality, reducing the path cost to 19, by adjusting its frontier evaluation based on forward-looking risk boundaries. A* expanded the most nodes (32) and required a higher memory footprint (51) but guaranteed safety and economic efficiency, producing the optimal solution with the lowest path cost of 16 while steering completely clear of all hazards. These results demonstrate that while GBFS variants are ideal for rapid, resource-constrained execution, A* remains necessary when path quality and absolute risk avoidance are the primary objectives.